CineMood

Multilingual Movie Review Sentiment Analysis

Member 2: Models and Evaluation

Project details	Value
Languages	English and Hindi
Classifiers	Naive Bayes; SMO (WEKA)
Evaluation	Stratified 10-fold cross-validation; seed 42
Software	WEKA 3.8.7; Java 17
Report date	1 November 2026

Scope: This report documents the completed Member 2 experiments using the cleaned ARFF datasets prepared by Member 1. Results and model file sizes are taken from the executed WEKA output shared for this project.

1. Objective and responsibilities

The objective is to classify movie reviews as negative or positive in English and Hindi, and compare Naive Bayes with Sequential Minimal Optimization (SMO) using WEKA. Member 2 is responsible for the classifier configurations, evaluation, saved model artifacts, and interpretation. Dataset acquisition, cleaning, labeling and export are Member 1 responsibilities.

2. Datasets and validation

Property	English	Hindi
ARFF relation	cinemood_en	cinemood_hi
Cleaned instances	1,993	2,227
Negative instances	996	946
Positive instances	997	1,281
ARFF file	en_reviews.arff	hi_reviews.arff
Approximate file size	7.0 MB	5.4 MB

Both files contain precisely two attributes: review (WEKA string) and sentiment (nominal: {negative,positive}); the class attribute is last. The Hindi ARFF preserves Hindi review text in Devanagari script. Both files were successfully parsed in the Java-dependent WEKA validation tests.

Member 1 reported 2,000 raw English records, reduced to 1,993 after removing 5 near-duplicates and 2 exact duplicates. For Hindi, 3,100 raw records were reduced to 2,227 by excluding 867 neutral-labeled items and 6 exact duplicates. These are preparation statistics supplied by the script, not independently re-audited here.

Validation: 15 automated tests passed, including two WEKA-specific checks of ARFF parsing, schema and Unicode preservation, after WEKA_JAR was configured.

3. Experimental methodology

Four experiments were run independently: English/Naive Bayes, English/SMO, Hindi/Naive Bayes and Hindi/SMO. Each used WEKA FilteredClassifier, with the unsupervised attribute filter StringToWordVector applied to the first attribute, followed by the specified classifier. The filter configuration was -R first -W 1000 -L (first attribute, vocabulary limit setting 1000, lowercase tokens). Other filter and classifier parameters were left at their WEKA 3.8.7 defaults.

Evaluation used stratified 10-fold cross-validation with seed 42, using the last attribute as the class. Encapsulating the text filter within FilteredClassifier ensures that vocabulary extraction is fitted inside the training portion of each fold rather than on all reviews before validation. No review metadata fields were supplied as predictors in these ARFF files.

The report uses only the Stratified cross-validation section for performance claims. The separate training-set statistics are not unbiased estimates of generalization.

4. Overall cross-validation results

Language	Classifier	Accuracy	Weighted precision	Weighted recall	Weighted F1
English	Naive Bayes	80.43%	0.805	0.804	0.804
English	SMO	78.17%	0.782	0.782	0.782
Hindi	Naive Bayes	63.36%	0.632	0.634	0.605
Hindi	SMO	69.96%	0.698	0.700	0.699

5. Class-wise cross-validation results

Experiment	Class	Precision	Recall	F1
English / Naive Bayes	Negative	0.793	0.823	0.808
	Positive	0.816	0.785	0.801
English / SMO	Negative	0.776	0.791	0.784
	Positive	0.787	0.772	0.780
Hindi / Naive Bayes	Negative	0.628	0.336	0.438
	Positive	0.635	0.853	0.728
Hindi / SMO	Negative	0.651	0.630	0.641
	Positive	0.733	0.751	0.742

6. Confusion matrices

Rows indicate the actual class and columns the predicted class. All four matrices below come from stratified 10-fold cross-validation, not the training-set matrices.

6.1 English - Naive Bayes (1,603 correct; 390 incorrect)

Actual / Predicted	Negative	Positive
Negative	820	176
Positive	214	783

6.2 English - SMO (1,558 correct; 435 incorrect)

Actual / Predicted	Negative	Positive
Negative	788	208
Positive	227	770

6.3 Hindi - Naive Bayes (1,411 correct; 816 incorrect)

Actual / Predicted	Negative	Positive
Negative	318	628
Positive	188	1093

6.4 Hindi - SMO (1,558 correct; 669 incorrect)

Actual / Predicted	Negative	Positive
Negative	596	350
Positive	319	962

7. Interpretation and findings

English: Naive Bayes obtained 80.43% accuracy and weighted F1 of 0.804, exceeding SMO (78.17%, F1 0.782) by 2.26 percentage points in accuracy. Under these configurations, Naive Bayes is the better-performing of the two evaluated English classifiers.

Hindi: SMO obtained 69.96% accuracy and weighted F1 of 0.699, exceeding Naive Bayes (63.36%, F1 0.605) by 6.60 percentage points in accuracy. SMO improved negative-class recall from 0.336 to 0.630, though positive-class recall decreased from 0.853 to 0.751. The Hindi Naive Bayes confusion matrix shows many negative reviews classified as positive (628 of 946).

Cross-language caution: The English and Hindi corpora differ in source composition, size, class balance and review lengths. Their score difference cannot establish that one language is intrinsically harder, nor does it demonstrate transfer or generalization between languages. No cross-language train/test transfer experiment was performed.

8. Limitations

The results are restricted to two available datasets, two classifiers, one preprocessing setup and one cross-validation seed. The Hindi class distribution is imbalanced (1,281 positive versus 946 negative). No hyperparameter search, confidence intervals, significance testing, independent external test set, or explicit cross-language transfer evaluation was reported. Since review-level grouping information was not part of the two-attribute modeling ARFF, potential grouping by movie or author was not evaluated here. Results should not be presented as production-system performance.

9. Reproducibility and saved artifacts

The experiments were executed on macOS with Java 17 and WEKA 3.8.7. The following project structure was confirmed during the workflow. The parent directory contains the locally cloned CineMood repository and sibling results, models and configs folders.

Artifact	Location relative to WEKA-Movie-Sentiment/
Prepared datasets	CineMood/data/processed/en_reviews.arff; hi_reviews.arff
Full evaluation outputs	results/english_naive_bayes.txt; english_smo.txt; hindi_naive_bayes.txt; hindi_smo.txt
Model configurations	configs/experiment_commands.txt
Trained full-data models	models/english_naive_bayes.model; english_smo.model; hindi_naive_bayes.model; hindi_smo.model
Additional training logs	models/english_smo_training.log; hindi_naive_bayes_training.log; hindi_smo_training.log

9.1 Evaluation command template

Run from the CineMood directory after setting WEKA_JAR to the installed WEKA JAR. Replace the dataset path and classifier for each experiment. The complete commands are saved in configs/experiment_commands.txt.

```
java -Xmx4g -cp "$WEKA_JAR" weka.classifiers.meta.FilteredClassifier -t data/processed/en_reviews.arff -c last -x 10 -s 42 -F "weka.filters.unsupervised.attribute.StringToWordVector -R first -W 1000 -L" -W weka.classifiers.bayes.NaiveBayes
```

9.2 Full-dataset model artifacts

Each saved .model file is a FilteredClassifier trained using its entire language-specific dataset with the same filter and classifier options and the WEKA -no-cv -d options. These artifacts are intended for later loading and inference. They are not the fold-specific models used to compute the reported cross-validation scores.

9.3 Reproduction checklist

1. Prepare and validate the two ARFF datasets using the Member 1 CineMood preparation script. 2. Verify that review is the first string attribute and sentiment is the final nominal attribute. 3. Set WEKA_JAR and use Java 17 / WEKA 3.8.7. 4. Run each saved experiment command from the CineMood folder. 5. Read only the Stratified cross-validation section for reported scores. 6. Compare accuracy, per-class metrics and confusion matrices with this report.

10. Conclusion

All four planned WEKA experiments were executed and their outputs saved. With stratified 10-fold cross-validation (seed 42), Naive Bayes was stronger on the English dataset (80.43% accuracy), whereas SMO was stronger on the Hindi dataset (69.96% accuracy). The differences apply to these datasets and configurations only. The reproducible experiment commands, evaluation text files and four full-data model artifacts complete the Member 2 modeling and evaluation deliverables.

Evidence basis: terminal outputs and ARFF headers supplied during the CineMood workflow, including the four WEKA cross-validation reports, dataset-preparation summary, automated-test output and model file listings. Dataset provenance and licensing should be cited from Member 1 documentation in the combined submission.